Exp. No: 1
18/7/2026

Network commands used in Linux and windows

AIM:

TO Study and execute basic networking commands in windows and Linux operating Systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status and troubleshooting network-related issues.

WINDOWS COMMANDS:

(i) ipconfig

Displays
- IP address
- Subnet Mask
- Default Gateway

SAMPLE OUTPUT

```
IPv4 Address . . . . : 192.168.1.20
Subnet Mask . . . . : 255. 255. 255 . 0
Default Gateway . . . . : 192. 168. 1. 1
```

(ii) ipconfig / all

Shows
- MAC Address
- DHCP Server
- DNS Server
- Lease information
- Adapter Details

(iii)
```
ping google.com
```
(or)
```
ping 8.8.8.8
```

Purpose

checks whether the destination is reachable

SAMPLE OUTPUT:

```
Reply from 8.8.8.8
Packets: sent = 4
Received = 4
Lost = 0
```

(iv)
```
tracert google.com
```

Purpose

Displays every router (hop) between your computer and destination

SAMPLE OUTPUT:

```
1   192.168.1.1
2   ISP Router
3   Google Network
```

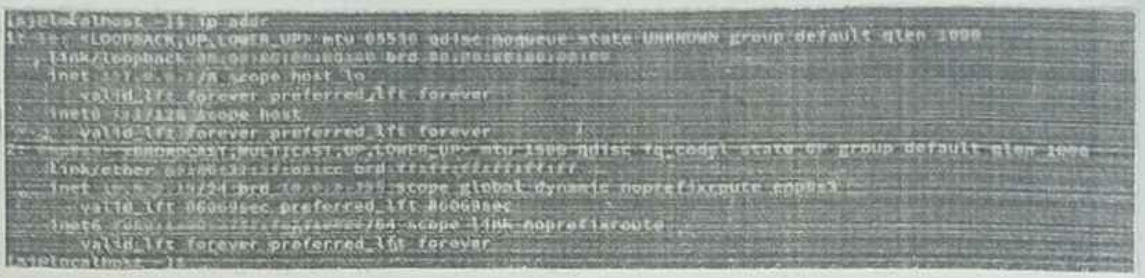
(v)
```
nslookup google.com
```

Purpose

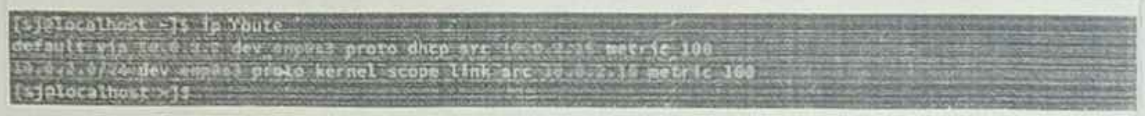
Displays
- DNS server
- IP Address of website

SAMPLE OUTPUT:

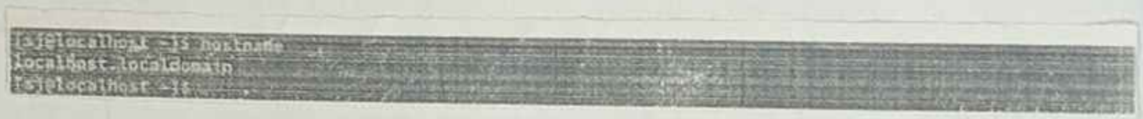
```
Name: google.com
Address: 142.250.xxx.xxx
```



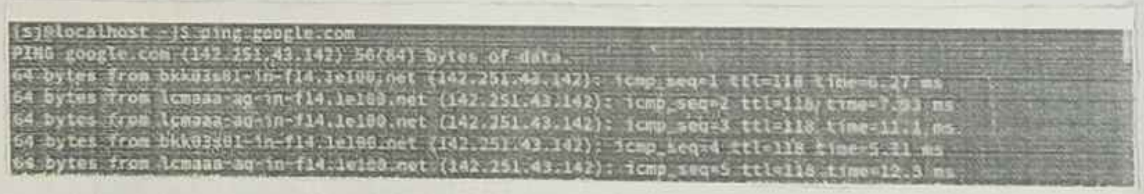
```
[sj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: [illegible]: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether [illegible] brd ff:ff:ff:ff:ff:ff
    inet [illegible].15/24 brd [illegible] scope global dynamic noprefixroute enp0s3
       valid_lft 86069sec preferred_lft 86069sec
    inet6 [illegible]/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[sj@localhost ~]$
```



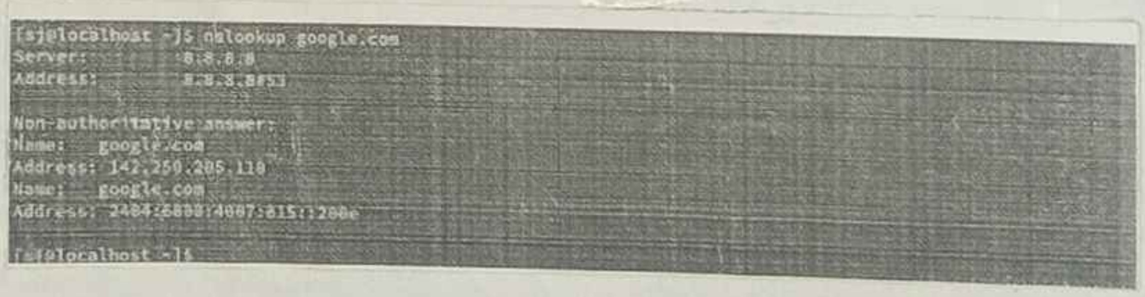
```
[sj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sj@localhost ~]$
```



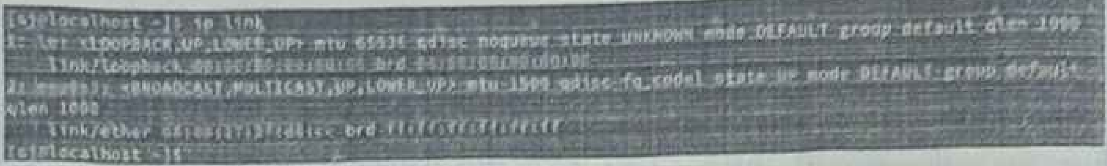
```
[sj@localhost ~]$ hostname
localhost.localdomain
[sj@localhost ~]$
```



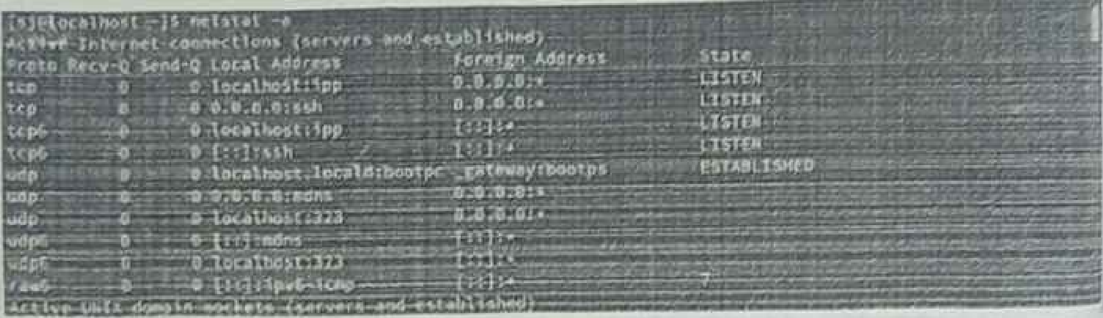
```
[sj@localhost ~]$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=118 time=6.27 ms
64 bytes from lcmaaa-ag-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=118 time=7.93 ms
64 bytes from lcmaaa-ag-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=118 time=11.1 ms
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=118 time=5.31 ms
64 bytes from lcmaaa-ag-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=118 time=12.3 ms
```



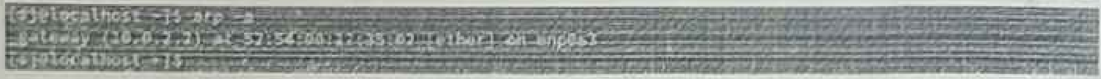
```
[sj@localhost ~]$ nslookup google.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4007:815::200e

[sj@localhost ~]$
```



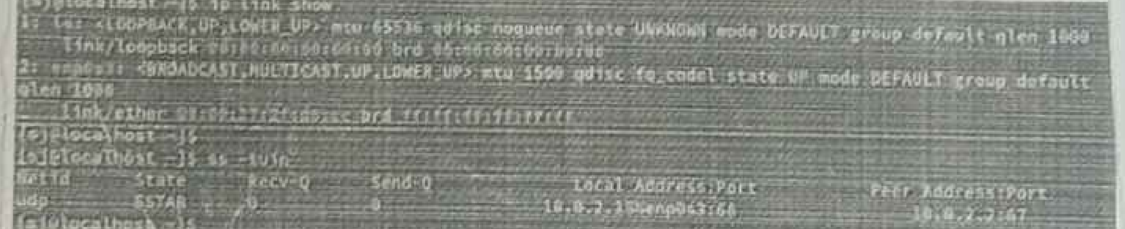
```
[s]@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether [illegible] brd ff:ff:ff:ff:ff:ff
[s]@localhost ~]$
```



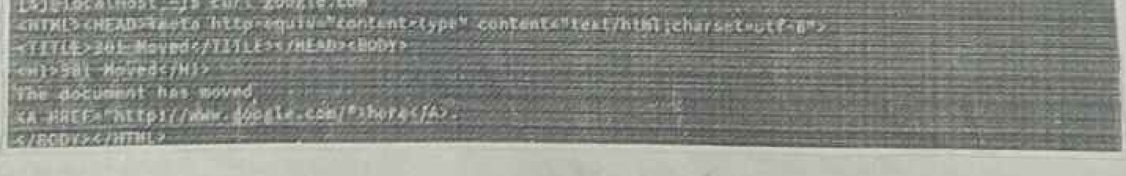
```
[s]@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
```



```
[s]@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[s]@localhost ~]$
```



```
[s]@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether [illegible] brd ff:ff:ff:ff:ff:ff
[s]@localhost ~]$
[s]@localhost ~]$ ss -tuln
Netid     State      Recv-Q      Send-Q             Local Address:Port            Peer Address:Port
udp       ESTAB      0           0               10.0.2.15%enp0s3:68                  10.0.2.2:67
[s]@localhost ~]$
```



```
[s]@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

(vi) route print

Purpose

Displays

* Network Routes
* Gateway
* Destination network

(vii) netstat

Purpose

Displays

* Active TCP connections
* Listening Ports

(viii) netstat -a

Shows

* All ports
* TCP connections
* UDP ports

(ix) getmac

Purpose

Displays MAC address of the computer

(x) hostname

Purpose

Displays computer name.

Example

DESKTOP - ABC123

(x) arp - a

Purpose

Displays IP-to-MAC address mappings

RESULT:

Basic networking commands were successfully executed on Linux and windows IP address, routing tables, ARP caches and DNS resolution were observed and recorded.